

No. of Functional Features Included in the Solution

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Feature Description	Module	Status	Marks Cont
1	Data ingestion	Load and inspect the loan dataset	Training	Done	1
2	Missing value handling	Fill categorical (mode) and numeric (mean) gaps	Training	Done	1
3	Categorical encoding	Map Gender, Married, Education, Self_Employed, Property_Area, Loan_Status to numeric codes	Training	Done	1
4	Class balancing	SMOTE oversampling of the minority class	Training	Done	1
5	Feature scaling	StandardScaler fit on the balanced feature set	Training	Done	1
6	Multi-model training	Train Decision Tree, Random Forest, KNN, XGBoost and compare accuracy/CV score	Training	Done	2
7	Model persistence	Save the best model (rdf.pkl) and scaler (scale1.pkl)	Training	Done	1
8	Web prediction form	Collect applicant details and return Approved/Rejected with confidence	Flask App	Done	2

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (preprocessing, balancing, scaling, model training, persistence, prediction form)
Additional / Nice-to-Have Features	2 (multi-model comparison UI copy, confidence score display)
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	home.html, predict.html, submit.html
2	Backend / Logic	2	app.py routing & form handling
3	Data / Model Storage	2	rdf.pkl, scale1.pkl
4	ML Pipeline	6	encoding, SMOTE, scaling, 4-model training, evaluation
5	Security / Authentication	0	Not yet implemented (planned future work)